

4E1330

Roll No. _____

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B.Tech. IV Sem. (Main/Back) Examination - 2023
Information Technology
4IT3-04 Principle of Communication

Time : 3 Hours**Maximum Marks : 70****Instructions to Candidates:**

Attempt all ten questions from Part A. Attempt any five questions out of Seven questions from Part B and three questions out of Five questions from Part C

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination (Mentioned in form No. 205)

Part - A

(word limit 25)

All questions are compulsory**(10×2=20)**

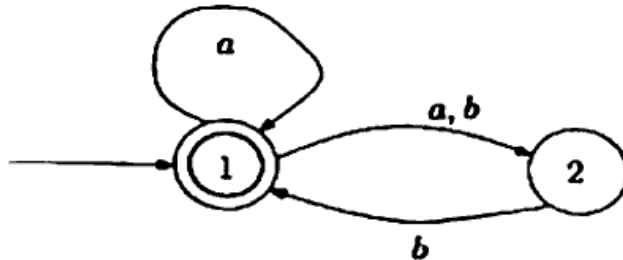
1. What is the over - modulation in AM?
2. Calculate the modulation index, m of an Amplitude modulation wave if its $V_{\max}(p-p) = 5.9$ divisions and $V_{\min}(p-p) = 1.2$ divisions.
3. What is the need of pre - emphasis circuit in FM?
4. Explain the flat - top sampling.
5. How PWM is different than PPM?
6. What do you mean by Quantization in PCM?
7. What is the Delta modulation?
8. Define the term 'pulse shaping'.
9. How QPSK is different than PSK modulation?
10. Define FHSS.

PART - B
(Word limit 100)

Attempt any Five questions.

(5×4=20)

1. Convert the following NFA to an equivalent DFA.



2. Design a Mealy Machine that computes 2's complement of the given binary input number.
3. Use the pumping lemma to prove that the following languages is not regular.
 $\{a^n b^n : n \geq 0\}$
4. Let G be the grammar $S \rightarrow 0B \mid 1A$, $A \rightarrow 0S \mid 1AA$, $B \rightarrow 1 \mid 1S \mid 0BB$. For the string 00110101, find (a) the leftmost derivation, (b) the rightmost derivation, and (c) the derivation tree.
5. Design a PDA accepting $\{a^n b^m c^n : m, n \geq 1\}$ by null store
6. Consider the TM description given M as shown in table. Draw the computation sequence of the input string 00.

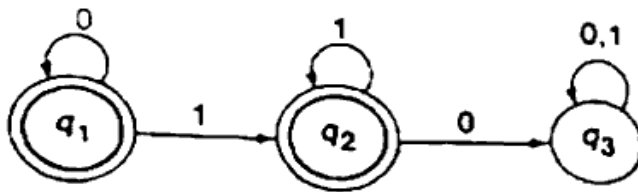
Present State	Tape symbol		
	b	0	1
$\rightarrow q_1$	1Lq ₂	0Rq ₁	
q ₂	bRq ₃	0Lq ₂	1Lq ₂
q ₃		bRq ₄	bRq ₅
q ₄	0Rq ₅	0Rq ₄	1Rq ₄
q ₅	0Lq ₂		

7. Write a note on Tractable and Untractable Problems.

PART - C

Attempt any **Three** questions.

1. Describe in English the set accepted by the finite automaton whose transition diagram is as shown in following figure. (3×10=30)



2. Consider the context-free grammar $G = (S, \Sigma, V, P)$, where $V = \{S, B\}$, $\Sigma = \{0, 1\}$, S is the start variable, and P consists of the rules $S \rightarrow BSB|B| \epsilon$, $B \rightarrow .00| \epsilon$. Convert this grammar to a context-free grammar in Chomsky normal form whose language is the same as that of G . Throughout the construction, upper case letters will denote variables.
3. Let $\Sigma = \{int, +, *, (,)\}$ and consider the language $ARITH = \{w \in \Sigma^* \mid w \text{ is a legal arithmetic expression}\}$. Design a PDA that accepts the $int + int * int$, $((int + int) * (int + int)) + (int)$ types of arithmetic expression?
4. Design a Turing machine over $\{1, b\}$ which can compute a concatenation function over $\Sigma = \{1\}$. If a pair of words $\{w_1, w_2\}$ is the input and the output has to be $w_1 w_2 \dots$
5. Explain the traveling salesperson problem? Why this problem is NP-complete?
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